

# Shahene Chaouachi

shahene@berkeley.edu | (415) 324-0494 | [LinkedIn](#) | [GitHub](#)

## EDUCATION

---

University of California, Berkeley | Berkeley, CA

May 2027

*Bachelor of Arts in Applied Mathematics*

- Relevant Coursework: Data Structures, Foundations of Data Science, Fundamentals of C++, Linear Algebra, Multivariable Calculus, Probability and Statistics, Numerical Analysis
- Activities: NASA Community College Aerospace Scholars Program, CS Club

## EXPERIENCE

---

*Hackathons@Berkeley – Berkeley, CA*

*Director - Sponsorships*

- Secured multi-tier sponsorships from top companies including Anthropic, Google Cloud, and Groq, supporting **Cal Hacks**, the largest collegiate hackathon (2,000 participants) and significantly expanding event resources.
- Built and maintained strategic sponsor **relationships**, driving long-term engagement and increasing sponsor retention across consecutive events.
- Led outreach campaigns and proposal development, achieving high commitment rates and strengthening the club's professional network.
- Coordinated cross-functional efforts with organizers and marketing teams to integrate sponsors into hackathon logistics and maximize visibility.

## PROJECTS

---

### Prepnest AI

- Led a two-person engineering team to design and develop an AI-powered web platform enabling IGCSE and A-Level students to convert static exam PDFs into interactive, auto-graded online assessments, enhancing accessibility and learning outcomes
- Integrated GPT-4 API for rubric-based grading and probabilistic scoring, achieving real-time feedback with sub-5 second response times and 95% alignment with official marking standards across validation datasets
- Built and maintained full-stack infrastructure using Node.js, Firebase Authentication, and Firestore database to support user management, data persistence, and scalability, serving 150+ active paying users
- Led backend performance optimization efforts, reducing system response latency by 28% and contributing to >99.9% uptime during pre-production testing and active user onboarding

### Movie Recommendation Platform

- Developed a semantic search and recommendation engine for 11,000+ movies using OpenAI's text-embedding-ada-002 API to encode movie plots and summaries, enabling deep contextual matching beyond keyword search
- Implemented efficient k-Nearest Neighbors similarity search with Faiss for scalable, low latency retrieval of semantically related movies
- Designed and built a modern, responsive React frontend inspired by Netflix with smooth animations, interactive carousels, and real-time search filtering to enhance user experience

## TECHNICAL SKILLS

---

**Languages:** Python, TypeScript, C++

**Frameworks/Libraries:** React, Node.js, Flask, TensorFlow, Pandas, NumPy

**Tools:** Git, PostgreSQL, Firebase